Distinct Values Sum

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array $x_1, x_2, \dots, x_n$. Let $d(a,b)$ denote the number of distinct values in the subarray $x_a, x_{a+1}, \dots, x_b$.

Your task is to calculate the sum $\sum_{a=1}^{n} \sum_{b=a}^{n} d(a,b)$, i.e., the sum of $d(a,b)$ for all subarrays.

Input

The first line has an integer n: the array size.

The next line has n integers $x_1, x_2, \dots, x_n$: the array contents.

Output

Print one integer: the required sum.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le x_i \le 10^9$

Example

Input	Output
5 1 2 3 1 1	29

Explanation: In this array, 6 subarrays have 1 distinct value, 4 subarrays have 2 distinct values and 5 subarrays have 3 distinct values. Thus, the sum is $6 \cdot 1 + 4 \cdot 2 + 5 \cdot 3 = 29$.